DataBase Lab 09 (IN-Lab Task)

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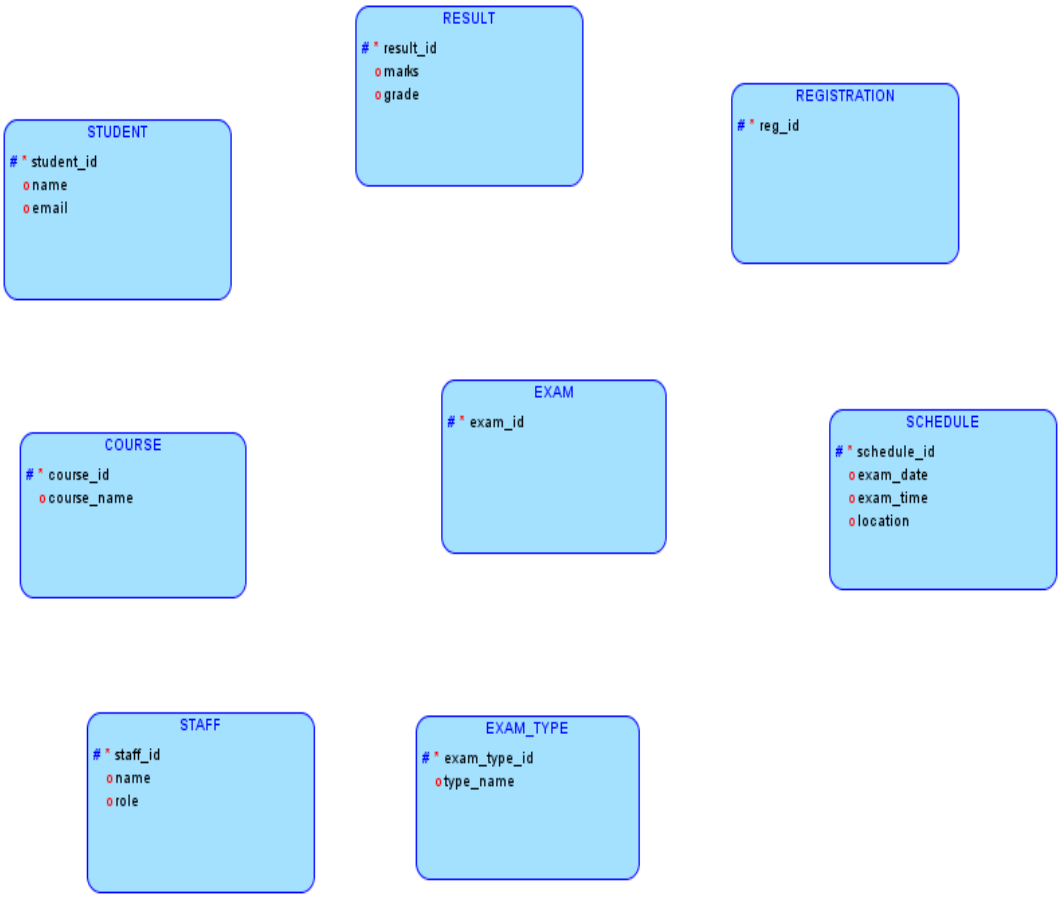
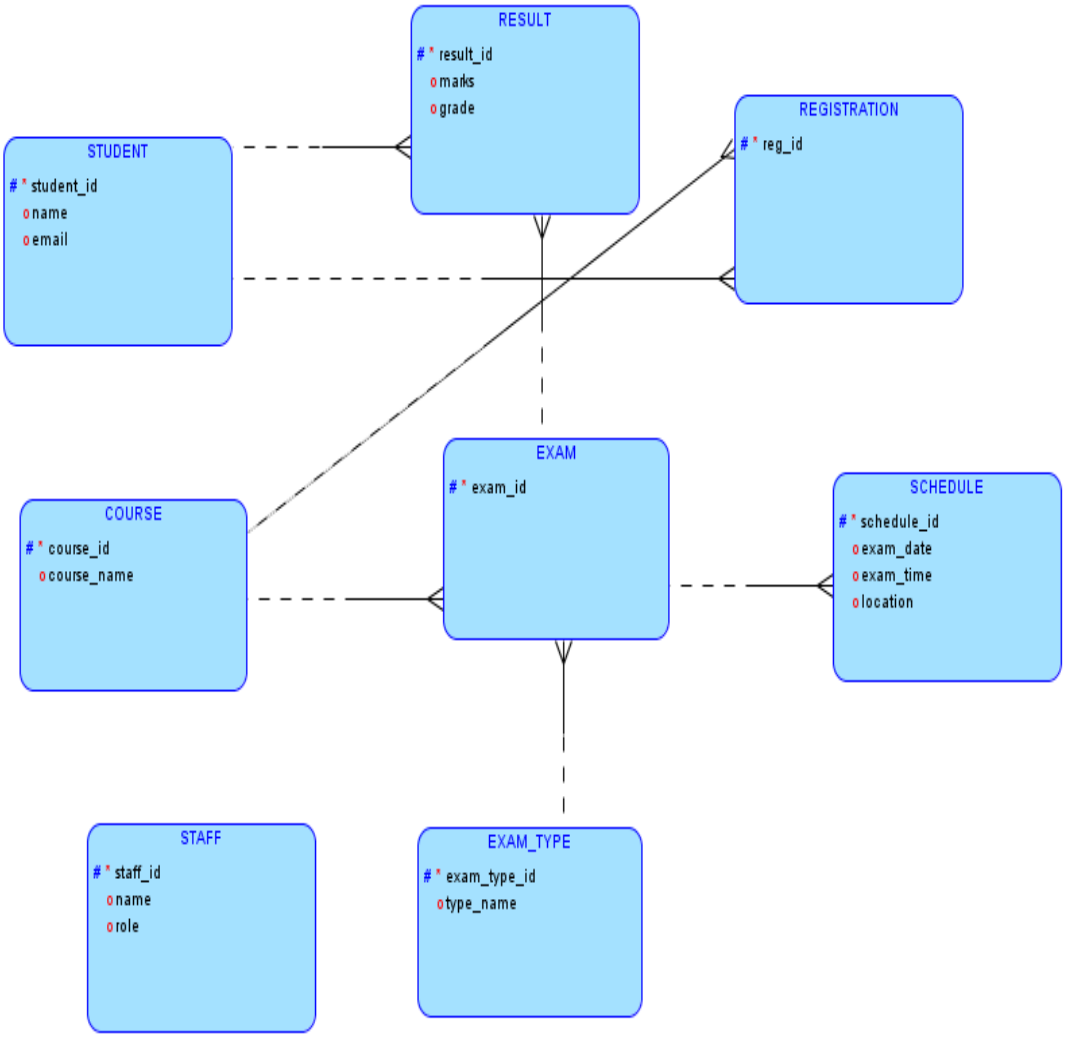
24k-0535

Examination Management System

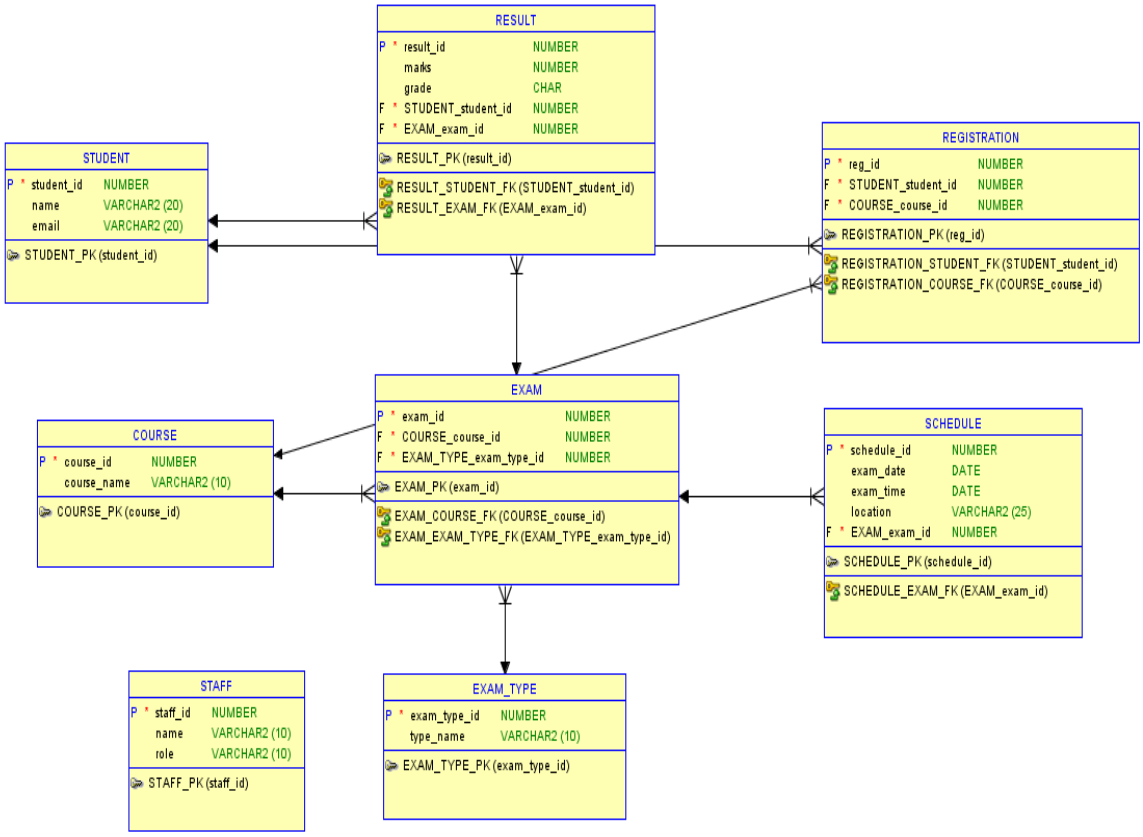
CREATING ENTITIES



ADDING RELATIONSHIPS



CONVERT TO RELATIONAL MODEL



GENERATING DDL

DDL File Editor - Oracle Database 11g

Oracle Database 11g | Relational_1 | Generate | Clear

```
-- Generated by Oracle SQL Developer Data Modeler 18.1.0.082.1041
--   at:        2026-04-04 15:23:34 PDT
--   site:      Oracle Database 11g
--   type:      Oracle Database 11g



CREATE TABLE course (
    course_id     NUMBER NOT NULL,
    course_name   VARCHAR2(10)
);

ALTER TABLE course ADD CONSTRAINT course_pk PRIMARY KEY ( course_id );

CREATE TABLE exam (
    exam_id                  NUMBER NOT NULL,
    course_course_id         NUMBER NOT NULL,
    exam_type_exam_type_id   NUMBER NOT NULL
);

ALTER TABLE exam ADD CONSTRAINT exam_pk PRIMARY KEY ( exam_id );

CREATE TABLE exam_type (
    exam_type_id   NUMBER NOT NULL,
    type_name      VARCHAR2(10)
);

ALTER TABLE exam_type ADD CONSTRAINT exam_type_pk PRIMARY KEY ( exam_type_id );

CREATE TABLE registration (
    reg_id               NUMBER NOT NULL,
    student_student_id   NUMBER NOT NULL,
    course_course_id     NUMBER NOT NULL
);

ALTER TABLE registration ADD CONSTRAINT registration_pk PRIMARY KEY ( reg_id );

CREATE TABLE result (
    result_id            NUMBER NOT NULL,
    marks                NUMBER,
    grade                CHAR
--  WARNING: CHAR size not specified
   ,
    student_student_id   NUMBER NOT NULL,
    exam_exam_id         NUMBER NOT NULL
);
```

```sql
ALTER TABLE result ADD CONSTRAINT result_pk PRIMARY KEY ( result_id );

CREATE TABLE schedule (
    schedule_id     NUMBER NOT NULL,
    exam_date       DATE,
    exam_time       DATE,
    location        VARCHAR2(25),
    exam_exam_id    NUMBER NOT NULL
);

ALTER TABLE schedule ADD CONSTRAINT schedule_pk PRIMARY KEY ( schedule_id );

CREATE TABLE staff (
    staff_id    NUMBER NOT NULL,
    name        VARCHAR2(10),
    role        VARCHAR2(10)
);

ALTER TABLE staff ADD CONSTRAINT staff_pk PRIMARY KEY ( staff_id );

CREATE TABLE student (
    student_id    NUMBER NOT NULL,
    name          VARCHAR2(20),
    email         VARCHAR2(20)
);

ALTER TABLE student ADD CONSTRAINT student_pk PRIMARY KEY ( student_id );

ALTER TABLE exam
    ADD CONSTRAINT exam_course_fk FOREIGN KEY ( course_course_id )
        REFERENCES course ( course_id );

ALTER TABLE exam
    ADD CONSTRAINT exam_exam_type_fk FOREIGN KEY ( exam_type_exam_type_id )
        REFERENCES exam_type ( exam_type_id );

ALTER TABLE registration
    ADD CONSTRAINT registration_course_fk FOREIGN KEY ( course_course_id )
        REFERENCES course ( course_id );

ALTER TABLE registration
    ADD CONSTRAINT registration_student_fk FOREIGN KEY ( student_student_id )
        REFERENCES student ( student_id );

ALTER TABLE result
    ADD CONSTRAINT result_exam_fk FOREIGN KEY ( exam_exam_id )
        REFERENCES exam ( exam_id );
```

```sql
ALTER TABLE result
    ADD CONSTRAINT result_student_fk FOREIGN KEY ( student_student_id )
        REFERENCES student ( student_id );

ALTER TABLE schedule
    ADD CONSTRAINT schedule_exam_fk FOREIGN KEY ( exam_exam_id )
        REFERENCES exam ( exam_id );



-- Oracle SQL Developer Data Modeler Summary Report:
--
-- CREATE TABLE                             8
-- CREATE INDEX                             0
-- ALTER TABLE                             15
-- CREATE VIEW                              0
-- ALTER VIEW                               0
-- CREATE PACKAGE                           0
-- CREATE PACKAGE BODY                      0
-- CREATE PROCEDURE                         0
-- CREATE FUNCTION                          0
-- CREATE TRIGGER                           0
-- ALTER TRIGGER                            0
-- CREATE COLLECTION TYPE                   0
-- CREATE STRUCTURED TYPE                   0
-- CREATE STRUCTURED TYPE BODY              0
-- CREATE CLUSTER                           0
-- CREATE CONTEXT                           0
-- CREATE DATABASE                          0
-- CREATE DIMENSION                         0
-- CREATE DIRECTORY                         0
-- CREATE DISK GROUP                        0
-- CREATE ROLE                              0
-- CREATE ROLLBACK SEGMENT                  0
-- CREATE SEQUENCE                          0
-- CREATE MATERIALIZED VIEW                 0
-- CREATE SYNONYM                           0
-- CREATE TABLESPACE                        0
-- CREATE USER                              0
--
-- DROP TABLESPACE                          0
-- DROP DATABASE                            0
--
-- REDACTION POLICY                         0
--
-- ORDS DROP SCHEMA                         0
-- ORDS ENABLE SCHEMA                       0
-- ORDS ENABLE OBJECT                       0
--
-- ERRORS                                   0
```